



First winter season of SARS-CoV-2, influenza and RSV monitoring in Belgian wastewaters

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In a nutshell

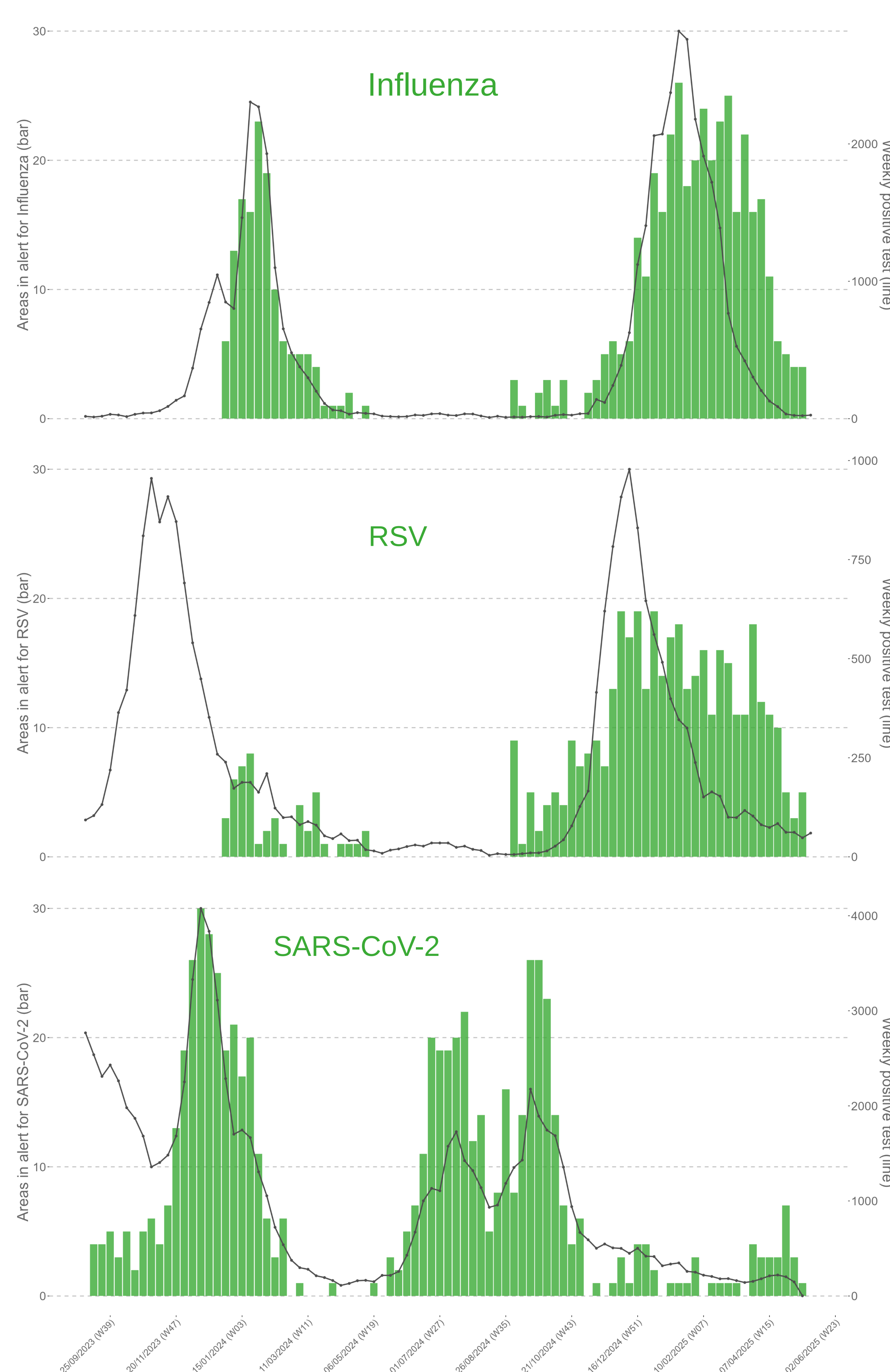
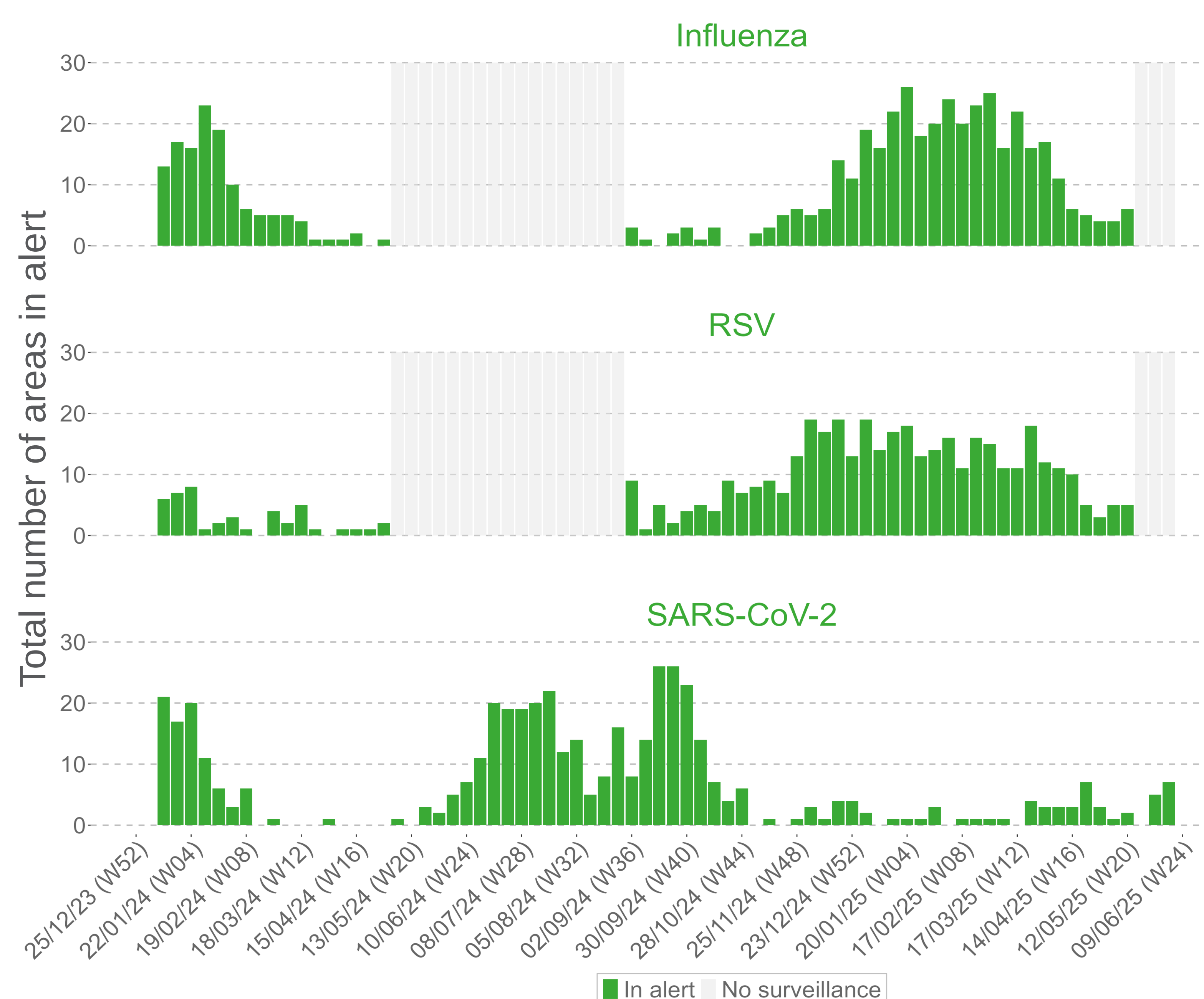
- 38% of the Belgian population monitored throughout 30 wastewater treatment plants
- Surveillance of SARS-CoV-2, influenza, RSV
- Significant correlation with clinical cases
- Added value: Large population covered, short reporting time, no data consolidation

Introduction

Wastewater-based epidemiology enables the development of cost-effective surveillance programs, compared to individual clinical tests for monitoring viruses circulation in a large population. In this work, we present the national wastewater-based epidemiology surveillance used by the public health authority to monitor SARS-CoV-2, RSV and influenza.

Methods

- Ultrafiltration with Centricon and RT-dPCR multiplex
- For SARS-CoV-2, an area is in alert if the threshold for the High Circulation indicator is reached. For RSV and influenza, an area is in alert if the virus is detected.
- Spearman correlation computed using R software (version 4.4.2)



REFERENCES

- SARS-CoV-2 Surveillance in Belgian Wastewater, Janssens et al., 2022, MDPI viruses
- Time series modelling for WBE of COVID-19, Bertels et al., 2023, STOTEN

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Results

- Lower concentration were measured for RSV and influenza than for SARS-CoV-2. Therefore, two distinct ways for reporting data were developed.
- High levels of correlation with clinical cases for influenza (0,84), RSV (0,59), and SARS (0,75).
- Added value: large population covered, short reporting time, no data consolidation
- Different population monitored for RSV through wastewater and clinical cases (children)
- Wastewater data are integrated to the existing surveillance system of respiratory viruses. Integrated reports are weekly communicated to the Risk Assessment Group and transferred in case of need to the Risk Management Group, who then decide of the relevant measures to be taken for Public Health.

Weekly report Available At



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